

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

**Course: B. Tech.
Semester :VII**

Branch : Electrical & Allied Engineering

**Subject Code & Name: BTEEC701 High Voltage Engineering
Max Marks: 60**

Date:02-01-2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain Poisson's equation in detail.	CO4	6
B) Explain the estimation and control of electric stress.	CO2	6
C) Explain breakdown in non-uniform fields and corona discharge.	CO4	6
Q.2 Solve Any Two of the following.		12
A) Explain Townsend's criterion for breakdown in electronegative gases.	CO1	6
B) Explain Streamers theory and Paschen's law in detail.	CO5	6
C) Describe current growth in presence of secondary processes.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) Explain the properties of solid, liquid and gases insulators	CO6	6
B) Explain the phenomenon treeing and tracking.	CO2	6
C) Describe propagation of lightning voltage & current waves on transmission lines.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Explain Lightning phenomenon in detail.	CO4	6
B) Describe multistage impulse generator and marx circuit.	CO2	6
C) Write a short note on equipment insulation level & co-ordination of substations	CO3	6
Q. 5 Solve Any Two of the following.		12
A) Explain impulse voltages Standard impulse waves shapes and it's equation	CO1	6
B) Explain I. E. C. & IS codes for high voltage tests on electrical appliances & power apparatus & electrical motors	CO2	6
C) Write a short note on generation of high d. c voltage.	CO6	6

*** End ***